

ASSIGNMENT - 2

Basic SQL Queries

1. CREATE TABLE

```
CREATE TABLE students ( id INT  
PRIMARY KEY AUTO_INCREMENT, name  
VARCHAR(100),  
age INT,  
department VARCHAR(50)  
);
```

2. INSERT - Add Data

```
INSERT INTO students (name, age, department)  
VALUES ('Geethika', 21, 'CSE');
```

```
INSERT INTO students (name, age, department)  
VALUES  
( 'Ravi', 22, 'ECE'),  
( 'Anita', 20, 'CSE');
```

3. SELECT - Retrieve Data

```
SELECT * FROM students;
```

```
SELECT name, department FROM students;
```

```
SELECT * FROM students  
WHERE department = 'CSE';
```

4. UPDATE - Modify Data

```
UPDATE students  
SET age = 23  
WHERE name = 'Ravi';
```

5. DELETE - Remove Data

```
DELETE FROM students  
WHERE id = 2;
```

6. ORDER BY (Sorting)

```
SELECT * FROM students  
ORDER BY age ASC;
```

```
SELECT * FROM students  
ORDER BY age DESC;
```

7. LIMIT - Top Records

```
SELECT * FROM students  
LIMIT 3;
```

8. Aggregate Functions

```
SELECT COUNT(*) FROM students;  
SELECT AVG(age) FROM students;  
SELECT MAX(age) FROM students;  
SELECT MIN(age) FROM students;  
SELECT SUM(age) FROM students;
```

9. GROUP BY - Used to group rows with the same values in a specified column and perform aggregate functions like COUNT(), SUM(), etc.

```
SELECT department, COUNT(*)  
FROM students  
GROUP BY department;
```

10. JOIN - Used to combine rows from two or more tables based on a related column between them.

```
SELECT students.name, departments.dept_name  
FROM students  
JOIN departments  
ON students.dept_id = departments.dept_id;
```

11. LIKE - Used to search for a specified pattern in a column (often with % wildcard).

```
SELECT * FROM students  
WHERE name LIKE 'G%';
```

12. ALTER TABLE - Used to modify the structure of an existing table (add, delete, or change columns).

```
ALTER TABLE students  
ADD email VARCHAR(100);
```

13. **DROP TABLE** - Used to permanently delete a table and all its data from the database.

DROP TABLE students;